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9. The work done by the Sun's gravitational force on the Earth is
   (a) always zero
   (b) always positive
   (c) can be positive or negative
   (d) always negative

10. If the mass and radius of the Earth are both doubled, then the acceleration due to gravity g'
   (a) remains same (b) $\frac{g}{2}$
   (c) 2 g (d) 4 g

11. The magnitude of the Sun's gravitational field as experienced by Earth is
   (a) same over the year
   (b) decreases in the month of January and increases in the month of July
   (c) decreases in the month of July and increases in the month of January
   (d) increases during day time and decreases during night time.

12. If a person moves from Chennai to Trichy, his weight
   (a) increases
   (b) decreases
   (c) remains same
   (d) increases and then decreases

13. An object of mass 10 kg is hanging on a spring scale which is attached to the roof of a lift. If the lift is in free fall, the reading in the spring scale is
   (a) 98 N (b) zero
   (c) 49 N (d) 9.8 N

14. If the acceleration due to gravity becomes 4 times its original value, then escape speed
   (a) remains same
   (b) 2 times of original value
   (c) becomes halved
   (d) 4 times of original value

15. The kinetic energy of the satellite orbiting around the Earth is
   (a) equal to potential energy
   (b) less than potential energy
   (c) greater than kinetic energy
   (e) zero

Answers

1) a	**2)** c	**3)** a	**4)** b	**5)** b
6) b	**7)** b	**8)** a	**9)** c	**10)** b
11) c	**12)** a	**13)** b	**14)** b	**15)** b

II. Short Answer Questions

1. State Kepler's three laws.
2. State Newton's Universal law of gravitation.
3. Will the angular momentum of a planet be conserved? Justify your answer.
4. Define the gravitational field. Give its unit.
5. What is meant by superposition of gravitational field?
6. Define gravitational potential energy.
7. Is potential energy the property of a single object? Justify.
8. Define gravitational potential.
9. What is the difference between gravitational potential and gravitational potential energy?
10. What is meant by escape speed in the case of the Earth?
11. Why is the energy of a satellite (or any other planet) negative?
12. What are geostationary and polar satellites?
13. Define weight